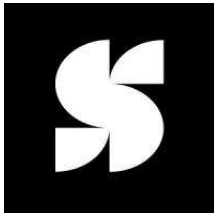


Are Your Job Candidates Cheating? Spotting Rehearsed and AI-Generated Answers in Real Time

Sponsor: Symbal AI

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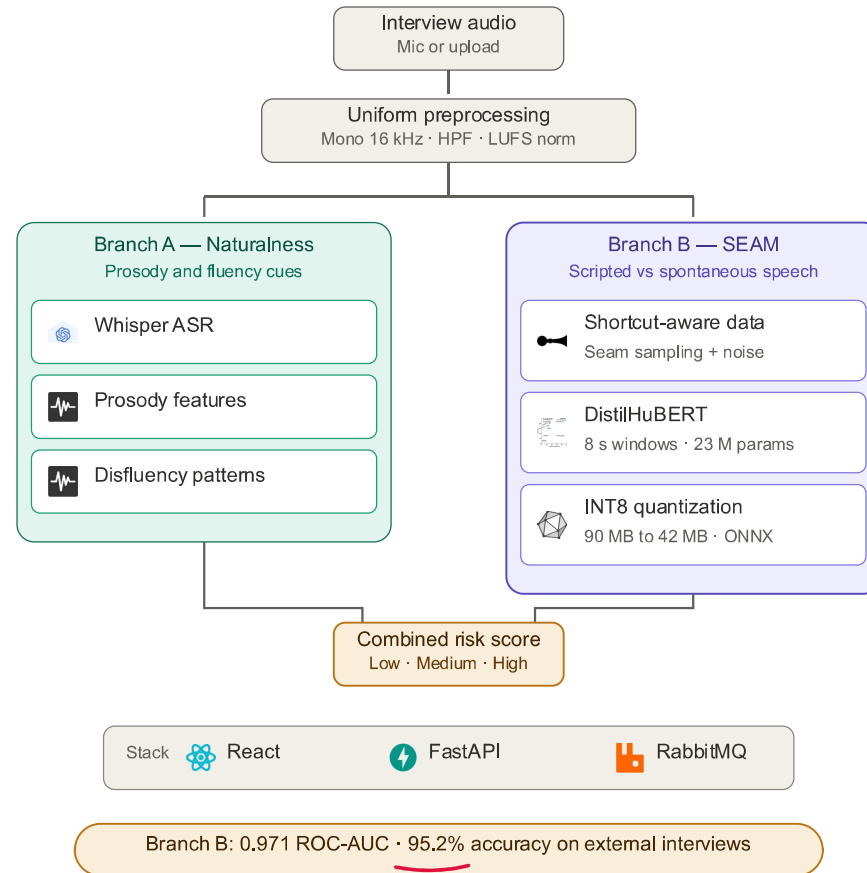


Abstract

- AI-assisted, asynchronous job interviews create a new integrity risk: **candidates reading scripted answers or piping in TTS-generated speech.**
- We built a real-time audio authenticity detector that **flags suspicious speech** as a narrow guardrail signal for human review, not a standalone decision.
- It uses a **dual-branch ML pipeline**, *React*, and containerized microservice backend.
- It is deployed to Symbal AI's test infrastructure on *Kubernetes* nodes with sub-second inference on CPU.

Features / Functionality

- Branch A detects unnatural speech by running *Whisper ASR* to extract word timing, then computing prosody features and disfluency patterns that distinguish spontaneous human answers from scripted or rehearsed delivery.
- Branch B identifies synthetic and scripted audio using a **compact trained model**.
- Both branches are **fused into a single calibrated risk score** per audio segment.
- The *React* frontend lets recruiters record directly from their microphone or upload an existing file, play back the audio, and **see per-branch and combined scores**.
- *FastAPI* backend **runs all inference locally on CPU** using *ONNX INT8* quantization, with **async RabbitMQ workers** enabling horizontal scaling across concurrent interview sessions.



Project Status / Future Work

- ✓ Deployed end-to-end system on Symbal AI servers
- ✓ Branch B research paper submitted to a **large speech conference**
- ✓ Model compressed for CPU inference; repo forked to Symbal's org
- ✓ *RabbitMQ* async integration for horizontal scaling
- Multilingual extension (**zero-shot transfer already above chance**)
- Broader device / environment calibration under distribution shift

Branch B — Research contribution

- The problem: benchmark scores on this task are easily inflated by shortcuts such as microphone, room acoustics, and corpus fingerprints, rather than actual speaking style.
- Addressed via uniform waveform preprocessing, seam-aware sampling (no cross-recording joins), a 14-hour non-speech noise bank, and a compact *DistilHuBERT* backbone (23 M params).
- Training on 240 h/class across four English corpora; evaluated on a held-out 720-clip interview benchmark.
- **0.971 ROC-AUC, 95.2% accuracy on external interview audio (3-seed mean).**
- Ablation confirms the claim: turning off shortcut defenses improves internal metrics but drops external ROC-AUC by ~17 points.
- *ONNX INT8* quantization: 368 MB to 41.8 MB with nearly no external accuracy loss.

Glossary

- **ASR**: Converts speech to text with word-level timestamps.
- **Disfluency**: Natural "um/uh" patterns common in human speech, rare in TTS.
- **TTS**: AI-generated synthetic speech; a key detection target.
- **ROC-AUC**: Performance metric; 1.0 = perfect, 0.5 is random chance.
- **ONNX INT8**: Compressed model format enabling fast CPU inference.

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